Experiment 9: Deploying ML Model as REST API with Cloud CI Integration

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AIM

To study deployment of a Machine Learning model as a REST API using Docker and cloud platforms with CI integration.

OBJECTIVES

1. To understand REST API-based model deployment.
2. To study offline deployment using Docker and Heroku CLI.
3. To understand online deployment using Azure or GCP endpoints.

THEORY

1. REST API-Based Model Deployment

In production, ML models are deployed as REST APIs so that external applications can send requests and receive predictions. This enables integration with web apps, mobile apps, and enterprise systems.

Containerization ensures the model and its dependencies run consistently across environments.

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2. Offline Setup: Docker and Heroku CLI

Docker is used to package the model API into a portable container image.

Using Heroku CLI, the containerized application can be tested locally before cloud deployment. This step ensures the API works correctly in a controlled environment and reduces deployment errors.

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3. Cloud Deployment Platforms

The container image can be hosted on platforms such as Heroku, DockerHub, or Render for public access.

For enterprise-grade deployment:

- Azure Machine Learning Online Endpoints
- Vertex AI (AI Endpoints)

These services provide scalable, secure, and managed model hosting.

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4. CI Integration and Workflow

With Continuous Integration (CI), model builds and deployments are automated whenever code changes occur.

The workflow includes building the Docker image, pushing it to a registry, and deploying it to cloud endpoints. This ensures consistent, automated, and reliable production deployment.

CONCLUSION

In this experiment, we studied deployment of an ML model as a REST API using Docker and cloud platforms. We also understood offline testing and online deployment using managed endpoints.

This approach ensures scalable, automated, and production-ready ML model serving.